



BASE24-atm[®]

IBM 3624 Device Support Manual

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Preface

The *BASE24-atm IBM 3624 Device Support Manual* provides an overview of the IBM 3624 ATM Device Handler process and IBM-related software options required by BASE24-atm. It also examines BASE24 support of the features and functions of IBM 3624 terminals.

Audience

ACI prepared this publication for distribution to institution personnel responsible for configuring and maintaining IBM 3624 terminals for use with BASE24-atm.

Prerequisites

This manual discusses a number of general topics relating to IBM 3624 terminals. These discussions have been included primarily to provide a stronger insight into the way BASE24-atm handles IBM 3624 terminals. This manual is not intended as a substitute for the vendor literature relating to IBM 3624 terminals. It is recommended that individuals making decisions on BASE24-atm IBM 3624 ATM processing be familiar with the vendor literature available for the type of IBM 3624 ATM they are using.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***BASE24 Device Control Manual*** contains detailed information on the operational control of the IBM 3624 terminals by BASE24. This includes instructions for such actions as downloading information and performing status inquiries.
- The ***BASE24 Base Files Maintenance Manual*** contains screen images and field descriptions of the Institution Definition File (IDF) and the External Message File (EMF).
- The ***BASE24-atm Files Maintenance Manual*** contains screen images and field descriptions of the Terminal Data File (TDF), Receipt File (RCPT), and the Transaction Log File (TLF).
- The ***BASE24-atm Settlement and Reporting Manual*** describes the settlement and cutover procedures, and contains samples and field descriptions of the reports produced by BASE24-atm.
- The ***BASE24-atm Transaction Processing Manual*** contains descriptions of the supported transactions and response codes, and the standard internal message (STM) used by BASE24.
- The ***BASE24 Logical Network Configuration File Manual*** describes the Logical Network Configuration File (LCONF) and its assigns and params.
- The ***BASE24 Transaction Security Manual*** contains information on transaction security options and message authentication.
- The ***BASE24 Text Command Reference Manual*** contains information on the commands that can be sent to the Device Handler process.
- Refer to the ***XPNET BASE24 Configuration Guidelines*** for more information on the Log Message Configuration File (LMCF) and setting up the source declarations for each terminal in the XPNET node.
- The ***BASE24 Log Message Reference Manual*** describes log messages, and how to access documentation for log messages.
- The ***BASE24 CRT Access Manual*** contains general information on logging on to the BASE24 system, using the default function keys, and creating screen access security.
- The ***BASE24 Tokens Manual*** describes the tokens that are added to the standard internal message to carry additional information.

Software Release

This manual documents standard processing for the BASE24-atm Release 5.1 product.

Customer-specific code modifications (that is, RPQs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” introduces the reader to the IBM 3624 ATM models that are supported by BASE24-atm and provides an overview of BASE24-atm support.

Section 2, “ATM Configuration,” contains information about IBM 3624 ATM configuration requirements, IBM 3624 load files, and the BASE24 files used in configuring IBM 3624 terminals.

Section 3, “BASE24-atm Device Handler Process,” contains an overview of the BASE24-atm IBM 3624 Device Handler process, the processing steps the Device Handler process takes when it is first started, an overview of the network commands that can be sent to the Device Handler process, and information on how to access the log messages generated by the Device Handler process.

Section 4, “ATM Printer Support,” describes BASE24-atm standard printer support for IBM 3624 terminals, along with configurable receipts as they apply to IBM 3624 terminals.

Appendix A, “IBM 3624 Terminal Servicing Procedures,” outlines onsite servicing procedures performed by administrative personnel at IBM 3624 terminals.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, AT-DA010-01 for the ***BASE24-atm IBM 3624 Device Support Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (for example, 08/96 for August, 1996) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions.

The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages.

Conventions Used in this Manual

This section explains a change in Nucleus process terminology.

Nucleus Process Terminology Change

There is one process that is the hub of an XPNET system. It manages message queues; controls all lines, stations, and processes in the system; and serves as an interprocess interface. Throughout this manual, this process is referred to as the XPNET process.

In other manuals, you may see references to the Nucleus process as performing these same functions. These two terms (Nucleus process and XPNET process) should be viewed as synonymous. The Nucleus process and the XPNET process perform the same functions and processing. As manuals are reissued, references to the Nucleus process will be replaced with references to the XPNET process.

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Section 1

Introduction

This section provides an introduction to BASE24-atm support of IBM 3624 terminals. It includes an overview of the following information:

- IBM 3624 ATM models supported by BASE24-atm
- IBM 3624 features supported by BASE24-atm and BASE24-atm support of protocols, transaction security, cutover, and settlement
- Financial and administrative transactions supported by BASE24-atm for IBM 3624 terminals
- Device configuration and control functions supported by BASE24

IBM 3624 Models

BASE24-atm supports the IBM 3624 version 7 and version 8 terminals. These full-service terminals provide cardholders the ability to perform routine banking transactions without directly contacting bank personnel. Each of the IBM 3624 terminals has four different models.

- Models 2 and 12 are designed to be installed through the wall of a building; thus, they were designed to withstand and detect penetration (such as cutting or drilling). Each is equipped with a heavy-duty front window, and a heavy-duty enclosure. Model 2 is a single dispenser terminal; it dispenses one denomination of bills. Model 12 is a dual dispenser terminal; it dispenses either one or two denominations of bills.
- Models 1 and 11 are designed to be installed in a protected area inside a building, such as a lobby. These models are not equipped with heavy-duty front windows or heavy-duty enclosures. Model 1 is a single dispenser terminal; it dispenses one denomination of bills. Model 11 is a dual dispenser terminal; it dispenses either one or two denominations of bills.

IBM 3624 version 7 terminals support PIN encryption and custom transactions and are equipped with low status indicators.

IBM 3624 version 8 terminals have the same functionality as IBM 3624 version 7 terminals, but they are also equipped with the multiline display feature, keyboard customization, and early identification card return. They also support multiple languages, fast cash transactions, transfers between accounts of the same types, PIN change at the terminal, surcharge fees, and the ability for the cardholder to complete a transaction without a receipt upon printer failure.

BASE24-atm Support

BASE24-atm software can process transactions originating at IBM 3624 terminals, as well as online control of the configuration, balancing, and functionality of each IBM 3624 terminal.

The BASE24-atm Device Handler process acts as the primary interface between BASE24 and an IBM 3624 terminal. The two primary types of support that the Device Handler process provides are control of the device itself, and control of the transactions that are processed by the device. Both types of support are discussed separately in the remaining pages of this section.

IBM 3624 terminals offer a variety of options, including risk-level options, PIN-handling techniques, and an auxiliary printer. Regardless of the options that are attached to the IBM 3624 terminals, BASE24 can be used as the processor for transactions originating at the ATM.

IBM 3624 version 8 terminals require two 2048-byte units of additional memory for standard BASE24 support.

Supported Features

The following IBM 3624 ATM features are supported by BASE24-atm:

- Receipt printer
- Journal printer
- Envelope depository
- DES PIN encryption
- Card capture
- Card reader
- Downloading of configuration files
- Multiple languages (version 8 only)
- Local PIN verification

Supported Protocols

BASE24-atm supports communications with IBM 3624 terminals using the following protocols:

- Bisynchronous
- SDLC XF
- SNAX/HLS
- Network Resident SSCP

Transaction Security

BASE24-atm offers transaction security options to protect an institution and its cardholders from fraud and includes such functions as PIN encryption and PIN verification.

BASE24-atm also supports the PIN change transaction for the IBM 3624 version 8 terminals, which enables the cardholder to perform a PIN change at the ATM. The PIN change transaction is not supported if local PIN verification is used.

BASE24-atm supports transaction security via both hardware and software. Hardware security options include interfaces to various models of Atalla and Racal-Guardata security hardware. The software security option is the DES method of PIN encryption and PIN verification.

The BASE24-atm IBM 3624 Device Handler process is responsible for the interface between BASE24 and the IBM 3624 terminals attached to it. In this capacity, the Device Handler process is responsible for PIN encryption between BASE24 and the IBM 3624 terminals. Based on a combination of device capabilities and your preferences, the Device Handler process passes on or decrypts PINs upon entry into the system.

For complete information on BASE24 transaction security options, refer to the ***BASE24 Transaction Security Manual***.

Local PIN Verification

BASE24-atm supports local PIN verification in which IBM 3624 devices that have PIN verification capability perform PIN verification at the device for financial and administrative transactions without the involvement of the BASE24 network.

To perform local PIN verification, the IBM 3624 terminal must be equipped with the proper hardware. Only those PINs whose offset is encoded on the card can be verified at the ATM.

You must consider which cards can be verified at the ATM because some interchanges do not support PINs that have been verified by the ATM.

BASE24-atm does not keep track of the number of bad PIN attempts whenever the ATM performs PIN verification.

Local PIN verification requires a setting in the BASE24-atm Terminal Data File (TDF) as well as changes to the version 7 CILOAD or version 8 CILOAD configuration files. The Customization Image Load File (CILOAD) must be modified in order for the IBM 3624 terminal to perform PIN verification. The parameters of the CFITST and CFIT macros in the CILOAD file define the characteristics of the PIN, specify whether PIN checking is to be performed by the terminal, and, if so, specify the characteristics of those data fields on the ATM card whose contents are used in the IBM 3624 PIN verification process. The CILOAD file must be recompiled and physically loaded or downloaded after these changes are made to implement local PIN verification. Refer to the appropriate IBM documentation for more information on setting up the configuration to support local PIN verification.

Cutover and Settlement

The role of BASE24-atm in settlement consists of providing you with detailed transaction reports that reflect cardholder and terminal activity on both a daily and monthly basis. BASE24-atm defines the business day, captures and keeps track of the transaction totals for each day, provides a daily file of transactions for processing by a host, and reports the funds due from and owed to the various participants in the logical network for balancing purposes.

A term used in connection with the BASE24-atm settlement process is cutover. Cutover refers to closing the current posting day and opening the next. It is the point in time when the transactions stop being logged to the current day's business and start being logged to the next day's business.

For a complete description of BASE24-atm settlement and cutover, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Transaction Support

BASE24-atm supports a number of financial and administrative transactions for IBM 3624 terminals.

Financial Transactions

The following financial transactions are supported by BASE24-atm for IBM 3624 terminals:

Note: Only IBM 3624 version 8 terminals can take advantage of multiple account select functionality, which allows cardholders to choose—from a list of their accounts—the account they want to use in a transaction.

- Withdrawal from checking account.
- Withdrawal from savings account.
- Cash advance from credit account.
- Fast cash (version 8 only). The withdrawal amount is predetermined, as is the account from which the cash is to be withdrawn.
- Deposit to checking account.
- Deposit to savings account.
- Transfer from checking to savings.
- Transfer from savings to checking.
- Transfer from credit account to checking.
- Transfer from checking to checking (version 8 only).
- Transfer from savings to savings (version 8 only).
- Payment enclosed.
- Payment from checking to credit account.
- Payment from savings to credit account.
- PIN change (version 8 only).
- Balance inquiry from checking.
- Balance inquiry from savings.
- Inquiry into credit account availability.

- Message from cardholder to financial institution.
- Log only 1.
- Log only 2.
- Log only 3.
- Log only 4.

Transaction Receipt Printing

All of the transactions listed above can have receipts printed for the cardholder's records. There are several ways that you can determine whether or not the receipt will print, and if so, how the receipt will look. Refer to section 4 for a further discussion of these options.

Optional Receipts

You can determine whether or not a receipt will be printed by transaction type. If you are using an IBM 3624 version 8 terminal, you can also allow the cardholder to determine if a transaction will be completed without a receipt if there is a printer failure.

Configurable Receipts

The BASE24-atm product comes with standard formats, or templates, for how financial and administrative receipts will look. You can use these templates or modify them to print the information that you want the cardholder to receive. Header templates allow you to customize the receipt to print your institution's name and marketing information.

PAN Truncation

You can determine if the account numbers on a receipt are printed in full or are truncated so only some digits of the number appear. This option allows you to reduce the amount of fraud caused by carelessly discarded receipts falling into the wrong hands.

Transaction Surcharge

An optional surcharge can be added to certain transactions on your terminals. This surcharge is in addition to any fees that the cardholder's institution imposes; in other words, surcharges are for transaction acquirers, not issuers. Surcharging is only supported on IBM 3624 version 8 terminals.

You determine whether the surcharge is added, and the amount added, depending on such variables as whether the transaction is on-us or not-on-us, the account type of the transaction, and the card type. Terminals can also be grouped for surcharging purposes. For example, all of your terminals in airports can have the same surcharge amount. You also determine whether or not a surcharge is assessed when a partial amount is dispensed and the rest of the transaction is reversed.

Surcharging is customized through the Terminal Data File (TDF), which is documented in the ***BASE24-atm Files Maintenance Manual***, and the Surcharge File (SURF), which is documented in the ***BASE24 Base Files Maintenance Manual***. The surcharge amount is carried in the Surcharge Data token which is appended to the Standard Internal Message (STM). Tokens are described in the ***BASE24-atm Tokens Manual***.

The following cardholder transactions can have a surcharge added by BASE24-atm:

- Withdrawal from checking account
- Withdrawal from savings account
- Fast cash withdrawal
- Cash advance from credit account

Administrative Transactions

You can perform administrative transactions in one of the following ways:

- You can use a machine-readable administrative card at the terminal.
- You can perform administrative services using the Device Control Terminal (DCT).

The administrative transactions available through BASE24-atm include the following transactions:

- Balance with current cash
- Balance with standard replenishment
- Terminal subtotals
- Increase cash

The terminal balancing transactions (that is, balance with current cash and balance with standard replenishment) allow terminals to be balanced using either the standard replenishment balancing method or the current cash balancing method. Totals stored in the Terminal Data File (TDF) are impacted and cleared as a result of these transactions.

The terminal subtotalling transaction allows activity totals, kept since the last balancing transaction up to the time the transaction is initiated, to be printed on audit and receipt printers. Terminal totals are not cleared with this transaction.

The terminal servicing transaction (that is, the midpoint cash adjustment transaction) allows the high or low bill denominations in a terminal to be replenished or increased without balancing the terminal. While the terminal is closed for servicing, the currency counters are cleared. Terminal servicing transactions that decrease the number of bills in the terminal cannot be made on IBM 3624 devices.

For more information on administrative transactions, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Device Configuration and Control

Prior to accepting transactions, an IBM 3624 ATM must be loaded or downloaded with its customization data. This customization data is contained in a CILOAD file which is built using an IBM program. Once built, this CILOAD file can be installed at the ATM or downloaded from BASE24 using the Device Control Terminal (DCT) functions. Additional information on the procedures used when downloading the CILOAD file is available in the ***BASE24 Device Control Manual***.

Terminal Balancing

Once a terminal is operational, it must be balanced on a regular basis. This can be accomplished via BASE24 DCT functions or administrative transactions at the ATM. The DCT allows a terminal to be closed, balanced, and opened again. Other functions, such as adding to the cash supply and servicing the terminal requires administrative transactions to be performed at the terminal site.

Procedures are provided in appendix A for performing various servicing functions at the ATM.

Log Messages

The BASE24-atm Device Handler process generates log messages periodically and as needed to notify you of the state of an ATM. The messages range in severity from informational to critical and can be printed or displayed on BASE24-atm logs. Refer to section 3 for information about how log messages are documented.

Terminal Control

The Device Control Terminal (DCT) is at the heart of the control mechanism for terminals within BASE24-atm. The DCT is the one set of screens that contains all of the functions necessary to control the terminal.

Basic device control functions common to all types of terminals include the following:

- Balancing terminals
- Inquiring into the status of terminals
- Opening and closing the terminals
- Starting and stopping the terminals
- Displaying subtotals
- Increasing cash totals in the terminal hoppers

Each of the functions listed above is associated with the DCT, since each is critical to the operation of the terminals. This central location allows these functions to be used more efficiently. For example, when you need to balance a terminal, you can close the terminal from the same screen from which you executed the balancing command.

BASE24-atm processes also can be reinitialized or warmbooted from the DCT. This is important because Device Handler processes must be warmbooted, or reinitialized, when changes that affect the Device Handler processes are made to the Logical Network Configuration File (LCONF) or the Receipt File (RCPT). Processes must also be warmbooted when holiday processing information has been changed in the Institution Definition File (IDF) record for the institution that owns the terminal.

Anyone responsible for controlling terminals should have access to the DCT. Refer to the **BASE24 CRT Access Manual** and the **BASE24 Device Control Manual** for more information on how to access and use the DCT.

Other ways to impact terminals within BASE24 include issuing commands via a network control facility. Refer to the **BASE24 Text Command Reference Manual** for more information about network control commands.

Section 2

ATM Configuration

BASE24 support of IBM 3624 terminals includes flexible configuration options. There are many different ways the IBM 3624 terminals can be configured, depending upon the your preferences. For example, you could allow some terminals to offer certain transactions, while terminals in other locations would not allow the same transactions. Also, you can change the wording of the different screen displays for terminals in different locations.

This section contains information about the Customization Image Load File (CILOAD) as well as the other BASE24 files used to configure IBM 3624 ATM processing.

IBM 3624 ATM Configuration

The IBM 3624 ATM is controlled by software that is stored in an IBM configuration file on the BASE24-atm system. This file, called the Customization Image Load File (CILOAD), is downloaded to the ATM using the Device Control Terminal (DCT).

In addition to configuring the internal software for each IBM device, a number of BASE24 files are used to configure transaction processing at the network level to support IBM 3624 terminals. The ATM directly accesses the CILOAD for processing information. BASE24 files provide indirect support to the ATM by defining the interaction between the Device Handler process and the ATM.

Because of the many combinations of options that are available during configuration, the configuration file and BASE24 files are discussed in this section one at a time. The various combinations that can be derived from these individual options are left up to you. The following files are used by the IBM 3624 terminals or their Device Handler processes:

- Customization Image Load File (CILOAD)
- Logical Network Configuration File (LCONF)
- Terminal Data File (TDF)
- Receipt File (RCPT)
- Network Environment Source File (NEFS)
- Institution Definition File (IDF)
- Log Message Configuration File (LMCF)

Customization Image Load File

A Customization Image Load File (CILOAD) must be present in IBM 3624 terminals before any financial or administrative transactions can be performed at the terminal. This CILOAD contains configuration information for the terminal and is downloaded to the terminal.

BASE24 supplies a sample CILOAD in a Tandem edit file, which can be edited on the Tandem. However, once edited, this edit source file must be placed on an IBM host, and a CIGEN program must be run against the file. The CIGEN program creates a CILOAD object file which can then be returned to the Tandem and downloaded to the ATM via the Device Control Terminal (DCT).

For complete information on the contents of the CILOAD and how to create it, refer to the appropriate IBM vendor documentation.

Logical Network Configuration File

The Logical Network Configuration File (LCONF) contains the information that links physical file names to file names as they are referenced in BASE24 code. The LCONF also contains information that sets processing values for an entire logical network.

The information that links physical file names to file names as they are referenced in code are known as assigns. The information that sets processing values are known as params. If a process name is associated with a param, then the param is used only by the process indicated in the param.

For information on accessing the LCONF, and complete information on individual assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual***.

Assigns

The following assigns are used by the Device Handler processes to locate the appropriate files:

ATM-DH-CILOAD — The fully qualified file name of the configuration file for IBM 3624 terminals (for example, B24.\$DATA.PRO1DATA.CILOAD). This assign is used by IBM 3624 Device Handler processes when downloading terminals and can be overridden by a corresponding value in the device-specific information in the individual TDF record of the ATM. In the case of IBM 3624 terminals, this assign must be present in the LCONF for all of the IBM 3624 Device Handler processes in a logical network.

ATM-RCPT — The physical file name of the Receipt File (RCPT).

HSF — The physical file name of the Hardware Status File (HSF).

IDF — The physical file name of the Institution Definition File (IDF).

LMCF — The physical file name of the Log Message Configuration File (LMCF). This file is used during the initialization and reinitialization of the Device Handler process.

SECURE-DEV1 — The name of the primary security module used for PIN management, message authentication, and security module key management.

SECURE-DEV2 — The name of the secondary security module used for PIN management, message authentication processing, and security module key management.

TDF — The physical file name of the Terminal Data File (TDF).

Params

The following params are used for setting processing values at the logical network level:

ATM-DH-AUTH-TIMER — The number of seconds the Device Handler process waits for a reply from the BASE24-atm Authorization process before denying a request.

ATM-DH-MAX-IN-CORE — The maximum amount of extended memory to allocate for TDF records locked on disk for transactions in progress.

ATM-DH-MAX-TDFS — The amount of memory required to build a table of TDF locations. This param represents the maximum number of TDF records associated with a single Device Handler process.

ATM-DH-OPT-LVL — The number of times the Device Handler process writes to the TDF for normal transactions.

ATM-DH-STALE-RQST-TIMEOUT — The number of seconds an incoming transaction can remain in the Device Handler process queue before being discarded by the Device Handler process as too old to process.

ATM-LN-CUTOVER — The time of day that the ATM logical network is cut over. The Device Handler process uses this time for Hardware Status Error Logging (ELARG) processing.

DATE-DISPLAY-TYPE — Indicates the format of the month and day portion of the dates on Device Handler process receipts (that is, DD/MM or MM/DD). The DATE-DISPLAY-TYPE param does not affect dates that include the year (that is, dates in the form YYMMDD).

SECURE-DEV-FAIL-NOTIFY — The interval at which messages are logged due to transactions that are denied because of a security device failure.

SECURE-DEV-TIM-LMT — The number of seconds to wait for a response from a security device.

SECURE-DEV-TYP — Specifies the type of security module being used by the logical network.

REVERSE-BAL-INQ — Specifies whether balance inquiry transactions can be reversed.

Terminal Data File

The Terminal Data File (TDF) record for an ATM defines much of its functionality. Whenever BASE24 receives a message from an ATM or sends a message to an ATM, it accesses the TDF record of the ATM. A separate record must exist in the TDF for every IBM 3624 ATM controlled by BASE24-atm.

A TDF record is accessed using BASE24-atm file maintenance functions. The TDF record is made up of twelve screens. The first six screens contain general information that is common to all terminals; the next three screens are specific to IBM 3624 ATM applications. For complete information on the values of the individual fields in the TDF and how the TDF can be maintained, refer to the ***BASE24-atm Files Maintenance Manual***.

General TDF Information

The basic information that is contained in every TDF record includes the following items:

- Reference information
- Transaction processing information
- Status information
- Transaction security information

Reference information includes information such as the physical location of the terminal, the owner of the terminal, and the Device Handler and Authorization processes that are involved in the transaction processing at the specified terminal.

Transaction processing information includes information on the kind of not-on-us transactions allowed at the terminal for a specific type of terminal. The Authorization process determines if a match exists between the sharing groups indicated in the TDF record for the ATM and the sharing groups set in the Institution Definition File (IDF) for the institution that issued the card accepted by the ATM. This check determines the type of transaction agreement, if any, that has been made between the ATM owner and the institution owning the card.

Transaction activity can be monitored through a series of timer settings established in the TDF for peak and nonpeak periods and holiday schedules. Receipt groups for configurable receipts are defined in the TDF.

Information on status changes to the ATM such as the state of the ATM and the amount of transaction activity for the ATM is also kept in the TDF.

Transaction security information relating to the encryption and decryption of data and local PIN verification is defined in the TDF.

Specific IBM 3624 Information in the TDF

The TDF also contains information specific to IBM 3624 terminals. Information that is particularly unique to IBM 3624 terminals concerns determining the level of risk the ATM is to assume when a hardware error is encountered. If the risk-level is exceeded, the ATM either closes or is put into service mode, based on a setting in the TDF.

Also the TDF record has a field which the Device Handler process uses to determine whether or not the ATM is to be configured to allow the cardholder to select the number of each denomination of bills to receive.

Finally, each TDF record has information such as the type of line protocol, the type of PIN encryption being used by the ATM, the fully qualified name of the CILOAD file for the ATM, as well as whether or not multiple languages are supported by the ATM.

Receipt File

The Receipt File (RCPT) is used to support the configurable receipt option, which provides you with the ability to establish ATM receipt formats that are unique to the institution. While the BASE24 configurable receipts option may be of limited use due to the limited space on an IBM 3624 receipt, this customization option supports the display of an informational or marketing message on the cardholder receipt, the use of flexible monetary symbols, and, for version 8 models, multiple languages. The configurable receipt option includes approved and denied financial transaction receipts as well as administrative receipts.

Receipt configuration is supported by device type, receipt group, transaction type, language code, card issuer FIID, *from* and *to* account type, and printer type. Receipt groups allow the use of a single receipt format for a group of ATM terminals rather than configuring receipts for an individual terminal. Receipt groups are specified for each terminal via screen 8 of the Terminal Data File (TDF). For further information on establishing receipt groups in the TDF, refer to the ***BASE24-atm Files Maintenance Manual***. The information on a receipt can also be tailored for specific card issuers; for instance, you can have different marketing information for on-us and not-on-us transaction receipts. Having different templates for different printers allows you to truncate the PAN on a receipt to reduce the chance of fraud, while allowing the entire PAN to be printed on the journal so that you have a complete record of the transaction. The receipt can also be customized to allow separate templates for different *from* and *to* account types, so for instance you can truncate savings account numbers differently than credit card numbers.

Note that there are several ways you can configure BASE24 to require receipts or allow transactions without a printed receipt. The first option allows you to decide by transaction type whether receipts are required, optional (that is, allow the transaction if the receipt printer is not operational), or not printed. These settings are on TDF screen 8. The second option allows the cardholder to decide if he or she wants to complete a transaction without a receipt if the printer is not operational, which overrides your settings on the TDF.

For a description of how the Device Handler process uses the RCPT in determining receipt wording and format as well as examples of default receipt templates for various transactions, refer to section 4.

Network Environment Source File

The Network Environment Source File (NEFS) is the source file used to create the Network Environment File (NEF). The Network Environment File (NEF) contains the source declarations for all lines, stations, processes, and devices in an XPNET node. These declarations define the network components to the system. Every IBM 3624 ATM in an XPNET node must have an entry in the NEF for the station and the line associated with the terminal. In addition, every IBM 3624 Device Handler process must have an entry in the NEF.

Detailed information about how to add IBM 3624 declarations to the NEFS is included in the *XPNET BASE24 Configuration Guidelines*.

Institution Definition File

The Institution Definition File (IDF) is used to define the processing parameters for each institution participating in the logical network. The IDF is read by the Device Handler process during an administrative transaction in order to retrieve the dates the Device Handler process requires for cutting over its terminals and the holiday information the Device Handler process uses for inactivity timing.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

Log Message Configuration File

The Log Message Configuration File (LMCF) contains one record for each log message that has been altered by a BASE24 network operator. Because LMCF records must be read into memory by the processes using them, updates to the LMCF do not take effect until the processes are reinitialized.

An LMCF record is accessed using BASE24 operator functions. The file itself contains changes made to log messages in the form of records. The routing code, severity, and text of each log message can be modified via BASE24 operator functions. Refer to the ***XPNET BASE24 Configuration Guidelines*** for more information on BASE24 operator functions and how to access the LMCF.

Section 3

BASE24-atm Device Handler Process

The Device Handler process is the primary interface between an ATM and BASE24. All messages that are sent by BASE24 to terminals must first be received by the Device Handler processes and then forwarded to the terminals. Likewise, all messages that are sent by terminals to BASE24 must first be received by the Device Handler processes.

Because each type of ATM has its own message format and way of processing transactions, separate Device Handler processes must exist for each type of terminal supported by BASE24. For example, IBM 3624 terminals must have their own type of Device Handler process, which differs from the Device Handler process used for Diebold 1000-series terminals or NCR 50XX/56XX-series terminals.

This section provides the following information:

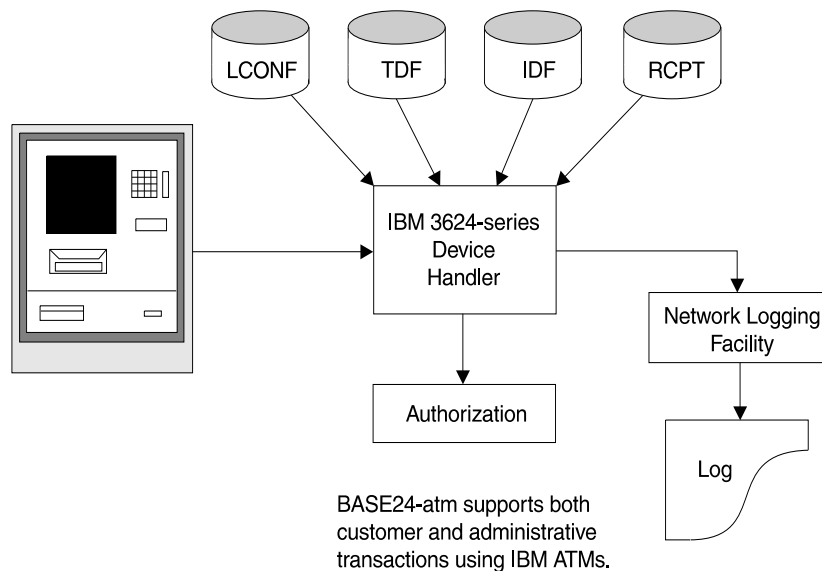
- An overview of the BASE24-atm IBM 3624 Device Handler process.
- Descriptions of the processing steps the Device Handler process takes when it is initialized and when it is warmbooted.
- Descriptions of BASE24 text commands the Device Handler process receives to perform specialized types of processing.
- Information on the log messages generated by the IBM 3624 Device Handler process.

BASE24-atm IBM 3624 Device Handler Process

IBM 3624 terminals are linked to BASE24-atm by a Device Handler process tailored to the IBM 3624 terminals. Every transaction initiated at an IBM 3624 terminal connected to BASE24-atm is passed to this Device Handler process when it enters the BASE24-atm system. The message coming from the terminal is in the native mode, and the Device Handler process converts this message into the BASE24-atm Standard Internal Message (STM) for sending to the BASE24-atm Authorization process. BASE24-atm supports both financial and administrative transactions via IBM 3624 terminals.

The Device Handler process creates log messages and sends them to the BASE24 logging facility when problems occur with the terminal. The logging facility then writes the log messages to the network log.

The following graphic illustrates the role of the IBM 3624 Device Handler process during transaction processing.



Transactions processed at different IBM 3624 terminals can be handled differently depending on how the individual terminal is configured in the Terminal Data File (TDF). Each terminal has a record in the TDF. This allows each terminal to have different sharing arrangements and transaction sets. The Device Handler process reads the TDF whenever it needs information about a terminal.

The Device Handler process uses the Institution Definition File (IDF) to obtain the dates the Device Handler process requires for cutting over its terminals and to obtain holiday information the Device Handler process uses for inactivity timing. It uses the Receipt File (RCPT) to determine the format and content of the receipt information that it sends to the terminals. And, it uses the Logical Network Configuration File (LCONF) to obtain the names of the TDF, IDF, and RCPT, plus a number of other file names and processing params.

Initialization Steps

The following steps are taken by the Device Handler process when it is first started or initialized.

1. Reads the name of the LCONF into memory.

The startup message from XPNET contains the name of the LCONF, which the Device Handler process reads into memory. This name allows the Device Handler process to access the LCONF for the assigns and params it uses in processing.

2. Reads the following assigns from the LCONF:

ATM-DH-CILOAD

IDF

LMCF

TDF

HSF

ATM-RCPT

3. Reads and edits the following params from the LCONF:

ATM-DH-AUTH-TIMER

ATM-DH-MAX-TDFS

ATM-DH-MAX-IN-CORE

ATM-DH-OPT-LVL

ATM-DH-STALE-RQST-TIMEOUT

ATM-LN-CUTOVER

REVERSE-BAL-INQ

DATE-DISPLAY-TYPE

4. Opens the TDF with read/write access using the name specified in the TDF assign in the LCONF.
5. Opens the RCPT with read access using the name specified in the ATM-RCPT assign in the LCONF.
6. Opens the IDF with read access using the name specified in the IDF assign in the LCONF.
7. Calls a security utility procedure that attempts to open the security device, if a security device is being used.

The security devices that are used by the network are specified in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF.

If the security devices cannot be opened, a log message is created and sent to the logging facility indicating that one or both of the security devices could not be opened.

The SECURE-DEV-FAIL-NOTIFY, SECURE-DEV-TIM-LMT, and SECURE-DEV-TYP params are also accessed in the LCONF at this time by the security utility procedure.

8. Initializes an extended memory area for finding and keeping TDF records.

The Device Handler process uses the ATM-DH-MAX-TDFS param in the LCONF to build a table of the TDF key record positions.

The Device Handler process uses the ATM-DH-MAX-IN-CORE param in the LCONF to determine how much memory is needed to keep copies of the TDF records in extended memory. This param indicates the maximum number of transactions that can be processed by a Device Handler process at one time.

9. Initializes an extended memory area for storing the IBM 3624-specific receipt records from the RCPT.
10. Determines if a Hardware Status Error Logging (ELARG) module is to be used (a separately purchaseable IBM module). If an ELARG module is to be used, the Device Handler process opens the HSF.
11. Initializes its timer table.

When a Device Handler process creates timers, they are placed in the Device Handler process timer table.

12. Initializes the TDF records, if applicable.
13. Builds holiday inactivity timer tables.
14. Adds intertransaction timers, if applicable.
15. Initializes the logging facility for writing log messages.
16. Reads the LMCF for its modified log messages.

The LMCF contains a record for each modified log message in the system. The Device Handler process opens the LMCF, retrieves its own modified log messages, and closes the file.

Warmboot Procedures

The warmboot command (9501****) is a general use operator command sent via the Device Control Terminal (DCT) screen or the network control facility. When Device Handler processes are warmbooted, they go through steps similar to their initialization routine. The steps they perform are described below:

1. Deallocates the holiday segment of the IDF.
2. Deallocates TDF segment.
3. Closes the TDF.
4. Closes the IDF.
5. Closes the HSF, if applicable.
6. Repeats steps 2 through 14 and step 16 of its initialization routine (described previously).
7. Logs a message indicating that warmboot is complete.

For information on the warmboot command and other operator commands, refer to the ***BASE24 Text Command Reference Manual***.

Device Handler Process Text Commands

Text commands are used to send information from a network control facility to a process. These commands serve as a tool that you can use to perform network management functions such as changing certain processing parameters, warmbooting a specific process, monitoring a specific process, and resolving problems encountered by a specific process.

The following text commands can be used with the BASE24-atm IBM 3624 Device Handler process:

- The OPEN command causes the Device Handler process to send the appropriate native-mode message to open the ATM.
- The CLOSE command causes the Device Handler process to send the appropriate native-mode message to close the ATM.
- The LOAD commands direct the Device Handler process to download some form of configuration information to the ATM.
- The RESET command causes the Device Handler process to interrupt any current transaction or load in progress for the given ATM. You should not enter this command unless requested to do so by ACI personnel.
- The warmboot command (9501****) directs the Device Handler process to warmboot. For a list of the steps the Device Handler process follows when a warmboot command is issued, refer to the topic “Warmboot Procedures” above.

For descriptions and the syntax of BASE24 text commands, refer to the ***BASE24 Text Command Reference Manual***.

Log Message Generation

The Device Handler process generates log messages to assist network operators in performing their daily tasks. Log messages provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

The Device Handler process generates log messages using a standard routing code of 1.

Log Message Documentation

Log message documentation is contained in files located in the BA51LOGM subvolume. Each file contains log messages for a specific BASE24 module. The files are organized according to product module ID (PMID). One PMID exists for each BASE24 module. The file for IBM 3624 version 7 Device Handler process messages is named AT24V7. The file for IBM 3624 version 8 Device Handler process messages is named AT3624.

For a complete explanation of log message documentation, refer to the ***BASE24 Log Message Reference Manual***.

Section 4

ATM Printer Support

Standard default receipt formats based on IBM guidelines are provided with BASE24-atm. You can customize these default receipt formats. This section provides examples of default financial and administrative receipt formats as well as instructions on how to configure customized receipts. Refer to the vendor's documentation for information on specifications for the physical properties and standard print formats supported by IBM.

BASE24-atm Receipts

BASE24-atm can produce a printed receipt for the cardholder, which provides a tangible result of the electronic transaction that just took place. Receipts are also printed on the journal printer, so that the ATM owner has a record of the transaction.

Types of Printed Receipts

BASE24-atm can print receipts for the following transaction types:

- Financial transactions
- Administrative transactions

Financial Transactions

Each financial transaction has its own type of receipt, so that the information provided to the cardholder is tailored to the requirements of the transaction. A denied transaction also has its own type of receipt. Whether a receipt is printed can be determined by the institution at the terminal level, or you can allow the cardholders to decide if they want to continue with transactions that would have no receipts printed because of a printer failure.

Administrative Transactions

Administrative transactions are performed by institution personnel to perform such tasks as replenishing currency. You can configure whether administrative receipts are printed in the TDF.

Configurable Receipts

Configurable receipts provide you with the ability to establish ATM receipt formats that are unique to the institution. In addition, this customization option supports the display of an informational or marketing message on the cardholder receipt, the use of flexible monetary symbols, and multiple language support (IBM 3624 version 8 only). The configurable receipt option allows for configuring approved and denied financial transaction receipts as well as administrative receipts.

Receipt configuration is supported by device type, receipt group, transaction type, language code, card issuer FIID, *from* and *to* account type, and printer type. Receipt groups allow the use of a single receipt format for a group of ATMs rather than configuring receipts for each ATM. Receipt groups are specified for each terminal via Terminal Data File (TDF) screen 8. For further information on establishing receipt groups in the TDF, refer to the ***BASE24-atm Files Maintenance Manual***. Receipts can be configured for each transaction in each language that the device supports. Customized receipt formats are configured by maintaining records in the Receipt File (RCPT).

IBM 3624 terminals print receipts on preprinted card stock, and each receipt can have four lines of 35 characters each of information actually printed by the terminal. While the information printed on the receipt is configurable, the size limitations and preprinted information of the receipt restrict what you can print on the receipt.

Receipt File

The Receipt File (RCPT) is the primary file used to configure receipts. It contains receipt templates that define the actual text and format to use for the various types of receipts.

During initialization, the Device Handler process reads the information in the RCPT into extended memory and then uses that information for creating receipts.

Note: Because the RCPT information is read into extended memory, the Device Handler process must be warmbooted every time the RCPT is modified for the modifications to take effect.

Initially, the RCPT contains default templates for each of the receipt types. You can use these default templates as they are, or you can modify the templates to print the information you want on the receipt.

Receipt Templates

Receipt templates are made up of a series of field markers and field type codes that define the information to be included on the receipt and the format of the receipt. Refer to the topics “Field Markers” and “Field Type Codes” for a description of field markers and field type codes that can be used in the receipt templates.

Individual receipt templates can be created for the various types of approved and denied financial transactions, and approved and denied administrative transactions. These same receipt templates can also be created in different languages (on IBM 3624 version 8 terminals only).

To determine the text and format to use for a receipt, the Device Handler process scans a template to identify the field markers and field type codes and their positions in the template. Then, based on the field type codes, the Device Handler process retrieves the required information and then formats the receipt information based on the field markers used and their positions in the template. Examples of the default templates provided in the RCPT for various transactions and examples of actual receipts created from these default templates are included later in this section.

Receipt templates are divided into two general categories, receipt header templates, and receipt body templates.

Receipt header templates control the information printed at the top of the receipt for all types of transactions. For IBM 3624 receipts, the header is the top line of the receipt.

Receipt body templates control the information printed on the remainder of the receipt; body templates are specific to the type of transaction (for example, withdrawal or deposit, approved or denied, financial or administrative). For IBM 3624 receipts, the body is the last three lines of the receipt.

To correctly format receipt information for a transaction, a Device Handler process must be able to retrieve the appropriate templates from its RCPT information.

Receipt Template Keys

Each receipt template is one record in the RCPT, and the key fields for RCPT records are as follows: device type, receipt type, receipt group, language code indicator, FIID, *from* account type, *to* account type, and printer type. This group of

key fields, described below, provides you with a great deal of flexibility in defining templates. For a comprehensive list of values that can be used in each of these key fields, refer to the ***BASE24-atm Files Maintenance Manual***.

Key Field	Description
DEVICE TYPE	Groups the RCPT records by type of device, allowing the Device Handler process to locate its records at initialization.
RECEIPT TYPE	Identifies the type of receipt template. Valid type codes have been defined by BASE24-atm.
RECEIPT GROUP	Allows the use of a single receipt template for a group of ATMs. Receipt groups are specified for each ATM on TDF screen 8. When selecting template data, the Device Handler process determines the group number assigned to an ATM in the TDF, and then searches for templates with the corresponding group. If an exact match cannot be found for the ATM group, the Device Handler process uses template records with asterisks in their group fields.
LANGUAGE CODE INDICATOR	Allows you to define the same receipt templates for more than one language. That is, a separate receipt template can be created in each of two languages (supported by IBM 3624 version 8 terminals only). A language indicator is sent to BASE24-atm from the ATM after the cardholder selects a preferred language for the transaction. The Device Handler process then uses this language indicator to select the templates to use. For example, you could configure one receipt template in English and one in Spanish in the RCPT. Each would be assigned a corresponding language indicator, which would allow the Device Handler process to select the appropriate template based on the language indicator sent from the ATM.
FIID	Allows a receipt to be formatted differently depending on who issued the card or owns the terminal. For instance, you may want different information printed for an on-us transaction than a not-on-us transaction, or you may want an abbreviated journal receipt for a terminal that you own.

Key Field	Description
FROM ACCOUNT TYPE	Allows a receipt to be formatted differently depending on the type of <i>from</i> account. For example, you can truncate savings account numbers differently than credit card numbers.
TO ACCOUNT TYPE	Allows a receipt to be formatted differently depending on the type of <i>to</i> account. For example, you can truncate savings account numbers differently than credit card numbers.
PRINTER TYPE	Allows a receipt to be formatted differently depending on the type of printer. For instance, you can truncate the account numbers on a financial receipt to help prevent fraud if the receipt falls into the wrong hands. However, you want the entire account number to print on the journal printer so that you have a complete record of the transaction.

Default Receipt Templates

Default receipt templates are provided in the RCPT for each type of transaction. These default receipt templates can be edited by you to customize specific types of information printed on the ATM receipts. Examples of the various default templates are described later in this section.

Multiple Languages

Receipt templates can be configured in two languages if you are using IBM 3624 version 8 terminals. That is, a separate receipt template can be created in each of the languages supported by each IBM terminal for a withdrawal transaction. The language indicator is sent to BASE24-atm from the ATM after the cardholder selects a preferred language for the transaction. For example, by configuring one receipt template in English and one in Spanish and matching each receipt template to the appropriate language indicator in the RCPT, cardholder receipts can be printed in English or in Spanish.

Components of the Receipt Template

A number of components make up a receipt template. They are as follows:

- Field markers
- Field type codes
- Static text
- Directives

Field Markers

Field markers are special characters that identify the type of data to be used on a receipt or specify an action such as a line feed. The field markers that can be used with financial and administrative configurable receipt templates in the RCPT are described on the following pages.

The following table describes the standard field markers that can be used in receipt templates to define receipt text and format.

Code	Description
#	Indicates that a variable data field is to be inserted. The number symbol (#) is immediately followed by a field type code which indicates the type of data to be inserted. The possible field type codes are discussed below in the topic “Field Type Codes.”
>	Indicates that a line feed character (hexadecimal 0A) is to be inserted.
@	Indicates that a literal data string will follow.
?	Indicates that a directive statement will follow. A directive gives a command other than printing. The possible directives are discussed below in the topic “Directives.”
^	Indicates the end of the print block. This character must occur at the end of each template except for the Header template (HDR).

Field Type Codes

Field type codes correspond to predetermined pieces of fixed-length data which the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the TDF, or the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals). The field type codes that can be used in transaction receipt templates in the RCPT are divided into codes used by financial transactions and codes used by administrative transactions. The field type codes are divided as follows:

- Standard field type codes. These codes are used with all receipt types.
- Cartridge field type codes. These codes are used with administrative receipts only.
- Terminal totals field type codes. These codes are used with administrative receipts only.

Data names are provided to indicate the specific data fields from which the information is taken. For further information on the STM data fields, refer to the ***BASE24-atm Transaction Processing Manual***. For further information on the TDF data names, refer to the ***BASE24-atm Files Maintenance Manual*** (using the data name index) or refer to the DDLs.

Static Text

You can enter any text in the receipt template that you want to print on the receipt. Static text must be preceded by the at symbol (@) in the template. You must ensure that the length of the static text plus the length of any field type codes used does not exceed the maximum number of characters that can be printed on the receipt.

Special characters that are reserved for use with field markers (#, >, @, and ^) can be printed in the text of the receipt by preceding them with the number symbol (#). For instance, to print the greater than character (>), you would enter #> in the template.

Directives

A directive indicates a command other than printing. The IF and ENDIF directives allow one template to print different text and/or data fields depending upon a condition that is determined outside of the RCPT. The IF directive marks the beginning of a block of receipt template that will be printed if the condition is true;

on the same line as the IF directive is the variable name of the condition to be tested. The ENDIF directive marks the end of the block. If the condition is not true, the block will be skipped, and the receipt will continue printing with the receipt line following the ENDIF directive.

Currently, FEES and NOFEES are the only valid condition values, and can only be used in financial receipt templates. When the FEES condition is on the same line as the IF directive, the following block will be printed if there is a surcharge fee assessed on the transaction. When the NOFEES condition is on the same line as the IF directive, the following block will be printed if there is no fee for the transaction (that is, surcharging is not configured, or surcharging is configured, but not applicable for this particular transaction).

An example of directives is described in the topics “Withdrawal from Checking, No Surcharge” and “Withdrawal from Checking, with Surcharge,” later in this section.

Financial Transaction Receipts

BASE24-atm supports both approved- and denied-transaction receipts. The following pages describe the default receipt templates that are used to configure approved-transaction receipts as well as denied-transaction receipts for financial transactions.

Default receipt templates are provided by ACI in English only regardless of the language indicator associated with the template. You must then modify the appropriate template if you want another language. For example, default receipt templates for language indicators 0 and 1 are configured in English. You must modify the templates to correspond to the languages supported. This section shows one example for each default receipt template in the RCPT.

The Device Handler process logs messages if receipt formatting errors occur when BASE24-atm transaction processing takes place. The Device Handler process also logs messages if the maximum number of lines per receipt or the print buffer is exceeded.

The ATM waits until each transaction is completed before printing the receipt. Because of IBM 3624 restrictions, BASE24-atm allows one transaction per receipt. After a cardholder completes the first transaction, the terminal can be configured to ask if another transaction is desired. When the cardholder wants to initiate another transaction, the terminal prints and dispenses the first receipt, and the cardholder is allowed to proceed. The second transaction is then printed on the next receipt.

Balance inquiry information requested by the cardholder can be printed, displayed on the ATM screen, or both, depending on how the TDF and IDF are configured.

Optional Receipts

You can determine whether receipts are printed for each transaction by several methods. The first method allows you to control receipt and audit printing by transaction type, and by whether the transaction was approved or denied. You enter this information on TDF screen 8, which is a device-specific screen. See the ***BASE24-atm Files Maintenance Manual*** for information on this file.

The second method allows the cardholder to make the final decision as to whether a transaction should be completed if a receipt cannot be printed. This method is only available if you are using IBM 3624 version 8 terminals. Note that the cardholder's choice with the second method overrides your settings on TDF screen 8 with the first method.

Configuring Optional Receipts

The optional receipts function is determined by settings on TDF screen 7, which is also a device-specific screen. The OPTIONAL RECEIPTS field on TDF screen 7 has two values, as follows:

- 0 = Optional receipts are not supported.
- 1 = The cardholder selects optional receipts if the printer fails. If the printer is working, the settings on TDF screen 8 are used.

Transaction Log File

The Transaction Log File (TLF) contains a flag indicating whether a receipt was requested. Valid values for the receipt option flag are as follows:

- 0 = The cardholder was not asked whether a receipt was required.
- 2 = A receipt was not printed because the printer has faulted.

You can view this flag on the TLF. Refer to the ***BASE24-atm Files Maintenance Manual*** for information on how to peruse the TLF. However, this flag is not on any of the printed BASE24 reports.

Sample Scenario

To ask the cardholder whether to continue with the transaction if the printer fails, enter the value 1 in the TDF field OPTIONAL RECEIPT.

If the printer is working, the settings on TDF screen 8 are used. If the printer faults the terminal is closed and reopened without the ability to print receipts. Each subsequent transaction is prefaced with the question, “Continue without a receipt?” If the cardholder selects yes, then all chained transactions are allowed without a receipt. If the cardholder selects no, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits the maintenance mode, the ability to print receipts is restored, and the “Continue without a receipt?” question is no longer asked of the cardholder.

Note that the text displayed to the cardholder on the terminal screen is determined by the CIRCPTS, which is a custom image load file. The BASE24 product does not provide a method for modifying this file.

Financial Transaction Field Type Codes

The following table describes the standard field type codes that are used in receipt templates. Each field type code corresponds to a predetermined piece of fixed-length data that the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the TDF, or the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals). Data names are included in parentheses after each field type code description to indicate the specific data field from which the information is taken. This list includes field type codes that are not used in the default receipt templates, but which you can use to customize the receipt templates.

Transaction amount fields are limited to a maximum of 10 characters including a monetary symbol (such as \$, ¥, or £), and decimal indicator. Balance amount fields are limited to a maximum of 13 characters including monetary symbol and decimal indicator. Amounts to be printed on receipts that exceed these limitations will cause blank fields on the transaction receipt. Refer to the topic “monetary Symbols” for more information on the symbols used to indicate currency type.

For further information on the STM data fields, refer to the **BASE24-atm Transaction Processing Manual**. For further information on the TDF data names, refer to the **BASE24-atm Files Maintenance Manual** (using the data name index) or refer to the DDLs.

Code	Description
A1	The transaction amount (TRAN-AMT field from the suspended transaction in the TDF device-dependent data), for instance, the withdrawal amount or deposit amount. In the case of a withdrawal with a surcharge fee, A1 contains the withdrawal amount plus the surcharge (which is the amount that will be deducted from the cardholder's account).
A2	An additional transaction amount (STM.RQST.AMT-2).
A3	An additional transaction amount (STM.RQST.AMT-3).
AB	The available balance (STM.RQST.AMT-3). This code does not print any descriptive text on the receipt. If this code is used on a balance inquiry receipt, the available balance is only printed if a response code 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields in the Institution Definition File (IDF) and the TDF indicate that Amount 3 (STM.RQST.AMT-3) is to be displayed.

Code	Description
CN	The card number and member number (TRACK2 field from the suspended transaction in the TDF device-dependent data) from the ATM card. To print card number only, see the CX field type code below.
CXl,r,c	The card number from the ATM card (TRACK2 field from the suspended transaction in the TDF device-dependent data). The member number is not printed. The three variables (<i>l</i>, <i>r</i>, and <i>c</i>) are used to control the truncation of the card number. If the variables are not present, the card number will be printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
DA	The transaction date. (STM.TRAN-DAT—MM/DD/YY or DD/MM/YY) The date displayed is dependent on the value specified in the DATE-DISPLAY-TYPE param in the Logical Network Configuration File (LCONF).
FA	The <i>from</i> account type, that is, CK, SV, CC. The account type printed on the receipt depends on the language indicator and the value of STM.FROM-ACCT-TYP. The Device Handler process matches account type numbers to a two-character account type description in the ACCT^TYP^TBL^G table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals).
FB	The 12 right-most digits of the <i>from</i> account number (RQST.FROM-ACCT field in the STM). If the RQST.FROM-ACCT field contains all blanks or zeros, the Device Handler process uses the account description from the ACCT^TBL table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals). The value in the RQST.FROM-ACCT-TYP field and the language indicator determine which table entry is used.
FD	The <i>from</i> account description, that is, CHECKING, SAVINGS, CREDIT CARD. The account description printed on the receipt depends on the language indicator and the value of STM.FROM-ACCT-TYP. The Device Handler process retrieves the account description in the APPL^TBL^G table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals).

Code	Description
FRL _{<i>l,r,c</i>}	The <i>from</i> account number (STM.RQST.FROM-ACCT). If the STM.RQST.FROM-ACCT is all blanks or zeros, the Device Handler process uses the account description from the APPL^TBL^G table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals). The value of STM.RQST.FROM-ACCT-TYP and the language indicator determine which table entry is used. The three variables (<i>l</i>, <i>r</i>, and <i>c</i>) are used to control the truncation of the account number. If the variables are not present, the account number will be printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
LB	The ledger balance (STM.RQST.AMT-2). This code does not print any descriptive text on the receipt. If this code is used on a balance inquiry receipt, the ledger balance is printed only if a response code of 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields in the IDF and the TDF indicate that Amount 2 (STM.RQST.AMT-2) is to be displayed.
MB	Cardholder member number. (STM.RQST.MBR.NUM)
OL	The number of the log only transaction selected (for example, Log Only #1.) Four Log Only transactions are supported. If the <i>from</i> account is 01, then a 1 is printed on receipt; if it is 02, a 2; if it is 03, a 3; and if it is 04, a 4.
RC	The response code. (STM.RQST.RESP)
RT	The response text. This code is used with denied transaction receipts. The printed receipt text depends on the value of STM.RQST.RESP. This response code is mapped to an internal response code specific to this Device Handler process. The Device Handler process searches the RESPONSE^TEXT^G table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals) to find a match on the internal response code. The corresponding text is then moved to the print buffer. The RESPONSE^TEXT^G table is configurable; refer to the topic “Response Code Text Table for Financial Receipts” later in this section.

Code	Description
S1	In the case of a withdrawal with a surcharge fee, this code is the amount to be dispensed, without including the surcharge amount. (STM.RQST.AMT-1 – TRAN-FEE)
S2	In the case of a withdrawal with a surcharge fee, this code is the amount of the surcharge (TRAN-FEE, from Surcharge Data token).
SQ	The 4-digit BASE24 sequence number assigned to the transaction. (TRAN-SEQ-NUM field from the suspended transaction in the TDF device-dependent data)
T1	The time the transaction took place—24-hour format HH:MM. (TRAN-TIM field from the suspended transaction in the TDF device-dependent data)
T2xx/yy	The time the transaction took place (TRAN-TIM field from the suspended transaction in the TDF device-dependent data). Time is displayed in a 12-hour format as opposed to the 24-hour format. If the time is less than 1200, the time appears as HH:MMxx. If the time is greater than or equal to 1200, the time appears as HH:MMyy where HH has valid values of 1 through 12. xx and yy are the configurable variable length symbols for a.m. and p.m., respectively. The format of this field type code is T2xx/yy. You can optionally insert spaces before xx and yy if you do not want the text to directly follow the time. yy is terminated by either the next field marker code (#, >, @, or ?), or by the end of the template.
TA	The <i>to</i> account type, that is, CK, SV, CC. The account type printed on the receipt depends on the language indicator and the value of STM.TO-ACCT-TYP. The Device Handler process matches account type numbers to a two-character account type description in the ACCT^TYP^TBL^G table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals).
TB	The 12 right-most digits of the <i>to</i> account number (RQST.TO-ACCT field in the STM). If the RQST.TO-ACCT field contains all blanks or zeros, the Device Handler process uses the account description from the ACCT^TBL table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals). The value in the RQST.TO-ACCT-TYP field and the multiple language indicator determine which table entry is used.
TC	The city where the terminal is located. (TDF.TERM-CITY)

Code	Description
TD	The <i>to</i> account description, that is, CHECKING, SAVINGS, CREDIT CARD. The account description printed on the receipt depends on the language indicator and the value of STM.TO-ACCT-TYP. The Device Handler process retrieves the account description in the APPL^TBL^G table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals).
TE	All 16 bytes of the terminal ID (TDF.TERM-ID).
TL	The terminal location (TDF.TERM-NAME-LOC).
TM	The first 8 bytes of the terminal ID (TDF.TERM-ID).
TN	The name of the terminal owner (TDF.TERM-OWNER-NAME).
TOL,r,c	The <i>to</i> account number (STM.RQST.TO-ACCT). If the STM.RQST.TO-ACCT is all blanks or zeros, the Device Handler process uses the account description from the APPL^TBL^G table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals). The value of STM.RQST.TO-ACCT-TYP and the language indicator determine which table entry is used. The three variables (<i>l</i>, <i>r</i>, and <i>c</i>) are used to control the truncation of the account number. If the variables are not present, the account number will be printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
TS	The state where the terminal is located. (TDF.TERM-ST-X)
TT	The transaction type. The transaction type printed on the receipt depends on the value of STM.TRAN-CDE. The Device Handler process searches the TRAN^TBL^G table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals) to find a match on the transaction code, then the corresponding text is moved to the print buffer.

Account Number Truncation

Three of the field type codes listed above have additional variables that allow the account number or card number to be truncated. This feature can help prevent fraud by making it more difficult for an unscrupulous person to use a carelessly discarded receipt to gain access to an account.

The three field type codes are CX (card number without member number), FR (*from* account number), and TO (*to* account number). The three variables that can be appended to these field type codes are *l*, a two-character numeric code which indicates the number of the left-most digits that are printed; *r*, a two-character numeric code which indicates the number of the right-most digits that are printed; and *c*, a one-alphanumeric character code which indicates the pad character, or the character used to fill the remaining digits. If these three variables are not present, the card or account number will be printed in its entirety.

Note that you can use a blank space as a pad character. To use a reserved special character, (#, @, >, and ^), precede it with the number symbol (#). For example, to print a > symbol, enter it in the template as #>.

You will need another receipt template with a different printer type if you want the card or account number to print in its entirety on the journal printer. Also note that if you do truncate one or more of the numbers, this will not affect how the number is stored internally; the complete number will be shown on the TLF and reports.

The following examples use this 19-digit primary account number (PAN):
1123581321345589144

Field Type Code in Receipt Template	PAN as Printed on Receipt
#CX00, 04, %	%%%%%%%%%%%%%%9144
#CX06, 04, #@	112358@@@@@@@@@9144
#CX06, 00, X	112358XXXXXXXXXXXXX

Monetary Symbols

The amount field type codes (A1, A2, A3, AB, LB, S1, and S2) can have a monetary symbol (such as \$, ¥, or £) that prints to the receipt. You can enter the monetary symbol either manually or have it determined automatically by the Device Handler process. To enter the symbol manually, type the appropriate symbol after each occurrence of the field type code. Any text after the code will be considered part of the monetary symbol until the next field marker is encountered.

The monetary symbol that is used can be determined automatically by the use of a table in the I3V7NAMS or I3V8NAMS. For the symbol to be determined automatically, you must enter CC\$ after the field type code. The symbol is determined by the currency code table (RCPT^MONEY^SYM^TBL^G) in the

I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals). This table contains an entry for each monetary system for which you want a special symbol to be printed. Each entry in the table contains the country code, the number of characters required in the sequence to send the special symbol(s) to the receipt printer (including escape codes), the number of characters taken up on the receipt as a result of the sequence, and the 10-character sequence that prints the proper monetary symbol. If a country code is encountered that is not in the table, the 3-character ISO currency code will be used (for example, USD for US dollars). If neither the country code nor the ISO currency code exist in the table, no symbol will be printed.

Transaction Receipt Header

The receipt header is printed at the top of the receipt for all financial transactions. The Device Handler process uses the header receipt template (the HDR receipt type in the RCPT) to format the receipt header for all transactions. As described earlier in this section, the process uses the device type, receipt group, transaction type, language code, card issuer FIID, from and to account type, and printer type from the RCPT to find the corresponding header receipt template.

Once this step is completed, the Device Handler process moves to the first character of the header receipt template and uses the field markers and field type codes described earlier in this section to scan the receipt template to determine the receipt text.

The following is a representation of the default entry in the header receipt template (receipt type HDR) in the RCPT for the Device Handler process.

TransactionType	Receipt Type	Default RCPT Template Data
General Header	HDR	#TM@ #DA@ #T2AM/PM@ #SQ>

Standard Receipt Samples

The following pages describe the default receipt templates provided in the RCPT. Default templates are provided in English only regardless of the language indicator associated with template. For example, default receipt templates for language indicators 0 and 1 are configured in English. You can modify the appropriate template if you want another language.

The following pages show default financial receipts for the following transactions:

- Withdrawal from checking, no surcharge
- Withdrawal from checking, with surcharge
- Deposit to savings
- Transfer from checking to savings
- Checking account inquiry
- Credit payment from checking
- Payment enclosed
- Message enclosed
- Log only
- PIN change
- Denied financial transactions

Withdrawal from Checking (IBM 3624 Version 7)

The receipt type WD is used for withdrawal transactions. The IBM default receipt template for IBM 3624 version 7 terminals is shown below, followed by a graphic example of the default receipt for a withdrawal from checking account transaction. Note that the IBM 3624 version terminal does not support the surcharge fee option. The account type can be printed on the receipt instead of the cardholder account number, depending on the authorization method used and whether the account number is available to BASE24-atm.

```
#CN@ #A1$>  
@ WITHDRAW #FB>  
@ #LB$@ #AB$^
```

TRANSACTION RECEIPT (preprinted information)			
ATM	MO/DAY/YR	TIME	SERIAL #
S1ASPARE	06/24/94	10:47 AM	801
CARD NUMBER			AMOUNT
1234567000000003-003			\$100.00
TRANSACTION		FROM ACCOUNT	TO ACCOUNT
WITHDRAW		013333	
\$82.75		\$82.75	
BANK IDENTIFICATION (preprinted information)			
1	2	3	
1234567890	1234567890	1234567890	1234

Withdrawal from Checking, No Surcharge (IBM 3624 Version 8)

The receipt type WD is used for withdrawal transactions. The IBM default receipt template for IBM 3624 version 8 terminals is shown below, followed by a graphic example of the default receipt for a withdrawal from checking account transaction that does not have an added surcharge fee. Note that in the case of a partial dispense, the actual amount dispensed is printed. The account type can be printed on the receipt instead of the cardholder account number, depending on the authorization method used and whether the account number is available to BASE24-atm.

```
?IF FEES
#CN@ #S1$>
?ENDIF
?IF NOFEES
#CN@ #A1$>
?ENDIF
@ WITHDRAW #FB>
?IF FEES
@FEE #S2$@ TOTAL AMOUNT#A1$^
?ENDIF
?IF NO FEES
@ #LB$@ #AB$^
?ENDIF
```

TRANSACTION RECEIPT (preprinted information)									
1	ATM S1ASPARE	MO/DAY/YR 06/24/94	TIME 10:47 AM						
	SERIAL # 801								
2	CARD NUMBER 1234567000000003-003		AMOUNT \$100.00						
3	<table border="1"> <thead> <tr> <th>TRANSACTION</th> <th>FROM ACCOUNT</th> <th>TO ACCOUNT</th> </tr> </thead> <tbody> <tr> <td>WITHDRAW</td> <td>013333</td> <td></td> </tr> </tbody> </table>			TRANSACTION	FROM ACCOUNT	TO ACCOUNT	WITHDRAW	013333	
TRANSACTION	FROM ACCOUNT	TO ACCOUNT							
WITHDRAW	013333								
4	\$82.75		\$82.75						
BANK IDENTIFICATION (preprinted information)									
1 2 3 1234567890123456789012345678901234									

Withdrawal from Checking, with Surcharge (IBM 3624 Version 8)

The following is a graphic example of the IBM 3624 version 8 default receipt format for a withdrawal from checking account transaction that has an added surcharge fee. Note that this receipt is produced using the same template as the example on the previous page. Because of the print restrictions of the IBM 3624, the balance information is replaced by the fee information.

TRANSACTION RECEIPT (preprinted information)			
ATM	MO/DAY/YR	TIME	SERIAL #
S1ASPRE	06/24/94	10:47 AM	801
CARD NUMBER			AMOUNT
1234567000000003-003			\$100.00
TRANSACTION	FROM ACCOUNT	TO ACCOUNT	
WITHDRAW	013333		
FEE	\$1.50	TOTAL AMOUNT	\$101.50
BANK IDENTIFICATION (preprinted information)			
1	2	3	
1234567890123456789012345678901234			

Deposit to Savings Account

The receipt type DP is used for deposit transactions. The IBM default receipt template is shown below, followed by a graphic example of the default receipt for a deposit to savings account transaction. The account type may be printed on the receipt, instead of the cardholder account number, depending on the authorization method used and whether the account number is available to BASE24-atm.

```
#CN@ #A1$>
@ DEPOSIT      #TB
@ #LB$@ #AB$^
```

TRANSACTION RECEIPT (preprinted information)			
1	ATM S1ASPARE	MO/DAY/YR 06/24/94	TIME 10:52 AM
	SERIAL # 802		
2	CARD NUMBER 1234567000000003-003		AMOUNT \$200.00
3	TRANSACTION DEPOSIT	FROM ACCOUNT 213333	TO ACCOUNT
4	\$482.75		\$482.75
BANK IDENTIFICATION (preprinted information)			
1 2 3 1234567890123456789012345678901234			

Transfer from Checking to Savings

The receipt type TR is used for transfer transactions. The IBM default receipt template is shown below, followed by a graphic example of the default receipt for a transfer from checking to savings transaction. Cardholder account numbers may be printed on the receipt, instead of the account type, depending on the authorization method used and whether the account number is available to BASE24-atm.

#CN@ #A1\$>
@ TRANSFER #FB#TB^

TRANSACTION RECEIPT (preprinted information)			
ATM	MO/DAY/YR	TIME	SERIAL #
S1ASPARE	06/24/94	10:55 AM	804
CARD NUMBER			AMOUNT
1234567000000003-003			\$200.00
TRANSACTION	FROM ACCOUNT	TO ACCOUNT	
TRANSFER	013333	213333	
BANK IDENTIFICATION (preprinted information)			

1 2 3
1234567890123456789012345678901234

Checking Account Inquiry

The receipt type INQ is used for inquiry transactions. The IBM default receipt template is shown below, followed by a graphic example of the default receipt for a balance inquiry from checking transaction. The cardholder account number may be printed on the receipt, instead of the account type, depending on the authorization level used and whether the account number is available to BASE24-atm.

```
#CN>
@ INQUIRY  #FB>
@  #LB$@   #AB$^
```

TRANSACTION RECEIPT (preprinted information)									
1	ATM S1ASPARE	MO/DAY/YR 06/24/94	TIME 10:55 AM						
	SERIAL # 804								
2	CARD NUMBER 1234567000000003-003		AMOUNT						
3	<table border="1"> <thead> <tr> <th>TRANSACTION</th> <th>FROM ACCOUNT</th> <th>TO ACCOUNT</th> </tr> </thead> <tbody> <tr> <td>INQUIRY</td> <td>013333</td> <td></td> </tr> </tbody> </table>			TRANSACTION	FROM ACCOUNT	TO ACCOUNT	INQUIRY	013333	
TRANSACTION	FROM ACCOUNT	TO ACCOUNT							
INQUIRY	013333								
4	\$384.47		\$384.47						
BANK IDENTIFICATION (preprinted information)									
1 2 3 1234567890123456789012345678901234									

Credit Account Payment from Checking Account

The receipt type PY is used for payment transactions. The IBM default receipt template is shown below, followed by a graphic example of the default receipt for a credit account payment made from a checking account. The cardholder account number may be printed on the receipt, instead of the account type, depending on the authorization method used and whether the account number is available to BASE24-atm.

```
#CN@ #A1$>
@ PAYMENT #FB#TB>
@ #LB@ #AB^
```

TRANSACTION RECEIPT (preprinted information)				
1	ATM	MO/DAY/YR	TIME	SERIAL #
	S1ASPARE	06/24/94	11:14 AM	810
2	CARD NUMBER		AMOUNT	
	1234567000000003-003		\$50.00	
3	TRANSACTION	FROM ACCOUNT	TO ACCOUNT	
	PAYMENT	013333	313333	
4	BANK IDENTIFICATION (preprinted information)			
	1	2	3	
	1234567890123456789012345678901234			

Payment Enclosed

The receipt type PYEN is used for payment enclosed transactions. The IBM default receipt template is shown below, followed by a graphic example of the payment enclosed transaction.

```
#CN@ #A1$>
@ PAYMENT ENC^
```

TRANSACTION RECEIPT (preprinted information)			
1	ATM S1ASPARE	MO/DAY/YR 06/24/94	TIME 11:14 AM
	SERIAL # 810		
2	CARD NUMBER 1234567000000003-003		AMOUNT \$50.00
3	TRANSACTION FROM ACCOUNT TO ACCOUNT PAYMENT ENC		
4			
BANK IDENTIFICATION (preprinted information)			
1 2 3 1234567890123456789012345678901234			

Message Enclosed

The receipt type MGEN is used for message enclosed transactions. The IBM default receipt template is shown below, followed by a graphic example of the default receipt for a message from the cardholder to the institution placed in the envelope depository.

#CN>
@ MSG ENC^

TRANSACTION RECEIPT (preprinted information)				
1	ATM S1ASPARE	MO/DAY/YR 06/24/94	TIME 11:14 AM	SERIAL # 810
2	CARD NUMBER 1234567000000003-003		AMOUNT	
3	TRANSACTION FROM ACCOUNT		TO ACCOUNT	
4	MSG ENC			
BANK IDENTIFICATION (preprinted information)				
1 2 3 1234567890123456789012345678901234				

Log Only

The receipt type OL is used for log only transactions. The IBM default receipt template is shown below, followed by a graphic example of the default receipt for a log only transaction.

```
#CN>  
@ LOGONLY #OL^
```

TRANSACTION RECEIPT (preprinted information)			
1	ATM S1ASPARE	MO/DAY/YR 06/24/94	TIME 11:30 AM
	SERIAL # 845		
2	CARD NUMBER 10010000000000-000		AMOUNT
3	TRANSACTION FROM ACCOUNT TO ACCOUNT LOGONLY 1		
4			
BANK IDENTIFICATION (preprinted information)			
1 2 3 1234567890123456789012345678901234			

PIN Change

The receipt type PC is used for PIN change transactions. The IBM default receipt template is shown below, followed by a graphic example of the default receipt for a PIN change transaction. Note that the PIN change transaction is available only on IBM 3624 version 8 terminals.

#CN>

@ PIN CHNG^

TRANSACTION RECEIPT (preprinted information)			
1	ATM S1ASPARE	MO/DAY/YR 06/24/94	TIME 11:14 AM
	SERIAL # 810		
2	CARD NUMBER 1234567000000003-003		AMOUNT
3	TRANSACTION FROM ACCOUNT TO ACCOUNT PIN CHNG		
4			
BANK IDENTIFICATION (preprinted information)			
1 2 3 1234567890123456789012345678901234			

Denied Financial Transactions

The Device Handler process converts receipt templates for denied transactions in the same manner as approved transactions with one exception. In addition to the receipt templates, a response code text table is used to associate one line of receipt text per ATM response code. Refer to the topic “Response Code Text Table for Financial Receipts” later in this section for a description of the response code text table.

The receipt type DENY is used for denied financial transactions. The IBM default receipt template is shown below, followed by a graphic example of the default receipt for a denied withdrawal transaction.

```
#CN>
#TT>
#RC@ #RT^
```

TRANSACTION RECEIPT (preprinted information)			
1	ATM	MO/DAY/YR	TIME
	S1ASPARE	06/24/94	11:14 AM
	CARD NUMBER		SERIAL #
2	1234567000000003-003		810
	AMOUNT		
3	51		
	TRANSFER		
4	51 EXPIRED CARD		
	BANK IDENTIFICATION (preprinted information)		
	1	2	3
	1234567890123456789012345678901234		

Response Code Text Table for Financial Receipts

The response code text is in two separate tables in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals) and is used in conjunction with the #RT (Response Text) data field when that particular data field is used in a receipt template for denied transactions.

The two tables used to supply text for the RT field type code are the RESPONSE^TEXT^TBL and REJECT^TEXT^TBL tables. The RESPONSE^TEXT^TBL table provides corresponding text for BASE24 response codes. The REJECT^TEXT^TBL table provides corresponding text for Device Handler process reject codes (these denote situations where the Device Handler process has rejected the transaction and not sent it any further for processing).

When the Device Handler process encounters an RT field type code in a receipt template, it supplies the text for the receipt from one of these tables in the I3V7NAMS or I3V8NAMS based on the response code or reject code associated with the transaction. For example, if BASE24 returned a response code of 55 to the Device Handler process, and #RT was specified in the receipt template, the default text that the Device Handler process would use is “TRANSACTION NOT ALLOWED.”

Refer to the *BASE24-atm Transaction Processing Manual* for complete descriptions of the BASE24-atm response codes.

Default Entries for Denied Transactions

Default entries for the response code text table are provided in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals).

The following table lists all of the default response code text messages in English. You can tailor all messages to meet your needs.

RC	Initial Message Text
50	UNAUTHORIZED CARD USAGE
51	EXPIRED CARD
52	INVALID CARD
53	INCORRECT PIN

RC	Initial Message Text
54	CONTACT BANK
55	TRANSACTION NOT ALLOWED
56	ACCOUNT NOT ALLOWED
57	TRANS NOT SUPPORTED
58	INSUFFICIENT FUNDS
59	INSUFFICIENT FUNDS W/BAL
60	USES LIMIT EXCEEDED
61	ENTER LESSER AMOUNT
62	PIN TRIES EXCEEDED
63	DAILY WD LIMIT REACHED
64	INVALID CREDIT CARD AMT
68	EXTERNAL DECLINE
69	NO SHARING W/ CRD ISSUER
70	SYSTEM ERROR
71	CONTACT CARD ISSUER
72	DESTINATION NOT AVAIL
73	ROUTING LOOKUP PROBLEM
74	MSG EDIT ERROR

Device Handler Process Reject Code Text

The following table provides, by reject code, the message text supplied with the initial software in the REJECT^TEXT^TBL table of the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals). In the initial version of the software, the same text is provided for both languages; however, you can modify this text as needed. Text can be a maximum of 24 characters.

RC	Initial Message Text
6	INVALID CARD
34	UNABLE TO PROCESS AMOUNT
47	UNABLE TO PROCESS
236	UNABLE TO PROCESS
237	INVALID TRAN SELECTION
241	TEMP. UNABLE TO PROCESS
246	EQUIPMENT ERROR

Administrative Transaction Receipts

The TDF accumulates transaction totals as transactions are completed for each ATM. These totals are used in balancing the ATM and monitoring the cash supply. They can also be printed at the ATM via terminal balancing and terminal subtotalling administrative transactions. All balancing totals are accumulated from the last time the ATM was balanced and the totals were reset to zero.

Administrative transaction receipts are printed on the same type of receipt paper forms as those used for financial receipts.

Administrative Transaction Field Type Codes

The following tables describe the standardized field type codes that are used in receipt templates for administrative receipts. Each field type code corresponds to a segment of fixed-length data. The data is retrieved from the TDF. Data names are shown in parentheses following each field type code description.

Cartridge Field Type Codes

The following table describes the cartridge field type codes that can be used in receipt templates for administrative receipts. Each cartridge field type code corresponds to a predetermined piece of fixed-length data that the Device Handler process retrieves from the Terminal Data File (TDF). Where applicable, TDF data names are included in parentheses after each field type code description to indicate the specific data field from which the information is taken.

In the default RCPT records for IBM 3624 terminals, cartridge field type codes are not followed by a monetary symbol to identify the monetary unit used for the transaction due to space restrictions on the receipt.

Totals for a specific cartridge are printed only if the bill value for that cartridge has been defined in the TDF.

Cartridge Field Type Code Descriptions	
Code	Description
HD	The number of bills removed from the cartridge by service personnel during the current balancing period multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[0].CASH-DECR field $\times$ the value in the TDF.HOPR[0].BILL-VAL field, where HOPR[0] refers to the cartridge containing the highest bill denomination). This field type code has a 10-byte length. Note: In the case of IBM 3624 terminals, cash decreases cannot be initiated using an administrative card; they can only be initiated via the Device Control Terminal (DCT).
HE	The current number of bills in the cartridge multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[0].CASH-END field $\times$ the value in the TDF.HOPR[0].BILL-VAL field, where HOPR[0] refers to the cartridge containing the highest bill denomination). This field type code has a 10-byte length.
HI	The number of bills added to the cartridge during terminal servicing for the current balancing period multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[0].CASH-INCR field $\times$ the value in the TDF.HOPR[0].BILL-VAL field, where HOPR[0] refers to the cartridge containing the highest bill denomination). This field type code has a 10-byte length.
HN	The number of bills dispensed from the cartridge for cardholder withdrawals during the current balancing period plus the number of bills removed from the cartridge by service personnel during the current balancing period multiplied by the denomination of the bills in the cartridge (that is, (the value in the TDF.HOPR[0].CASH-OUT field + the value in the TDF.HOPR[0].CASH-DECR field) $\times$ the value in the TDF.HOPR[0].BILL-VAL field, where HOPR[0] refers to the cartridge containing the highest bill denomination). This field type code has a 10-byte length.

Cartridge Field Type Code Descriptions	
Code	Description
HO	The number of bills dispensed from the cartridge for cardholder withdrawals during the current balancing period multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[0].CASH-OUT field $\times$ the value in the TDF.HOPR[0].BILL-VAL field, where HOPR[0] refers to the cartridge containing the highest bill denomination). This field type code has a 10-byte length.
HS	The number of bills in the cartridge at the beginning of the current balancing period for the terminal, multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[0].BEG-CASH field $\times$ the value in the TDF.HOPR[0].BILL-VAL field, where HOPR[0] refers to the cartridge containing the highest bill denomination). This field type code has a 10-byte length.
LD	The number of bills removed from the cartridge by service personnel during the current balancing period multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[1].CASH-DECR field $\times$ the value in the TDF.HOPR[1].BILL-VAL field, where HOPR[1] refers to the cartridge containing the lowest bill denomination). This field type code has a 10-byte length. Note: In the case of IBM 3624 terminals, cash decreases cannot be initiated using an administrative card; they can only be initiated via the Device Control Terminal (DCT).
LE	The current number of bills in the cartridge multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[1].CASH-END field $\times$ the value in the TDF.HOPR[1].BILL-VAL field, where HOPR[1] refers to the cartridge containing the lowest bill denomination). This field type code has a 10-byte length.
LI	The number of bills added to the cartridge during terminal servicing for the current balancing period multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[1].CASH-INCR field $\times$ the value in the TDF.HOPR[1].BILL-VAL field, where HOPR[1] refers to the cartridge containing the lowest bill denomination). This field type code has a 10-byte length.

Cartridge Field Type Code Descriptions	
Code	Description
LN	The number of bills dispensed from the cartridge for cardholder withdrawals during the current balancing period plus the number of bills removed from the cartridge by service personnel during the current balancing period multiplied by the denomination of the bills in the cartridge (that is, (the value in the TDF.HOPR[1].CASH-OUT field + the value in the TDF.HOPR[1].CASH-DECR field) $\times$ the value in the TDF.HOPR[1].BILL-VAL field, where HOPR[1] refers to the cartridge containing the lowest bill denomination). This field type code has a 10-byte length.
LO	The number of bills dispensed from the cartridge for cardholder withdrawals during the current balancing period multiplied by the denomination of bills in the cartridge (that is, the value in the TDF.HOPR[1].CASH-OUT field $\times$ the value in the TDF.HOPR[1].BILL-VAL field, where HOPR[1] refers to the cartridge containing the lowest bill denomination). This field type code has a 10-byte length.
LS	The number of bills in the cartridge at the beginning of the current balancing period for the terminal, multiplied by the denomination of the bills in the cartridge (that is, the value in the TDF.HOPR[1].BEG-CASH field $\times$ the value in the TDF.HOPR[1].BILL-VAL field, where HOPR[1] refers to the cartridge containing the lowest bill denomination). This field type code has a 10-byte length.

Terminal Totals Field Type Codes

The following table describes the terminal totals field type codes that can be used in receipt templates for administrative receipts. Each terminal totals field type code corresponds to a predetermined piece of fixed-length data that the Device

Handler process retrieves from the Terminal Data File (TDF). Where applicable, TDF data names are included in parentheses after each field type code description to indicate the specific data field from which the information is taken.

Terminal Totals Field Type Codes	
Code	Description
AC	The total amount of commercial deposits accepted since the terminal was last balanced (AMT-CMRCL-DEP field in the TDF).
AD	The unverified amount of deposits accepted since the terminal was last balanced (AMT-DEP field in the TDF).
AK	The unverified amount of checks cashed and received since the terminal was last balanced (AMT-CHK field in the TDF).
AP	The unverified amount of payments enclosed received since the terminal was last balanced (AMT-PAY field in the TDF).
CR	The total number of cards retained since the terminal was last balanced (CRDS-RET field in the TDF).
CK	The number of checks cashed since the terminal was last balanced (NUM-CHK field in the TDF).
D0	The number of deposits received since the terminal was last balanced (NUM-DEP field in the TDF).
D1	The number of commercial deposits received since the terminal was last balanced (NUM-CMRCL-DEP field in the TDF).
EN	The sum of the following: number of deposits, number of commercial deposits, number of message envelopes, number of payment enclosed envelopes, and number of checks cashed since the terminal was last balanced (that is, the sum of the following TDF fields: NUM-DEP, NUM-CMRCL-DEP, NUM-MSG, NUM-PAY, and NUM-CHK).
MG	The total number of “message to financial institution” envelopes received since the terminal was last balanced (NUM-MSG field in the TDF).
PY	The total number of payment enclosed envelopes received since the terminal was last balanced (NUM-PAY field in the TDF).

Administrative Templates

The following table shows the default receipt body templates used by IBM 3624 Device Handler processes for approved and denied administrative transactions. The same default template data is used for both language designations in the RCPT, so only one template is shown. The table also indicates the type of transactions for which the template is used as well as the receipt type used to identify the template in the RCPT.

Transaction Type	Receipt Type	Default RCPT Template Data
Approved administrative transactions	AD	@HB START#HS@ ADJ#HI@ OUT#HN> @LB START#LS@ ADJ#LI@ OUT#LN> @ENVELOPES #EN@ CARDS RETAINED #CR^
Denied administrative transactions	ADDN	> #RT> @ADMINISTRATIVE TRANSACTION DENIED^

Administrative Receipt Sample

The following is an example of an IBM ATM default receipt format for an administrative subtotal transaction. Note that this same format is used for all administrative transactions.

TRANSACTION RECEIPT (preprinted information)				
1	ATM	MO/DAY/YR	TIME	SERIAL #
	S1ASPARE	06/24/94	10:12 AM	64
2	CARD NUMBER		AMOUNT	
	HB START199940 ADJ		0 OUT 0	
3	TRANSACTION	FROM ACCOUNT	TO ACCOUNT	
	LB START	9740 ADJ	0 OUT 0	
4	ENVELOPES 1 CARDS RETAINED 0			
BANK IDENTIFICATION (preprinted information)				
1 2 3				
1234567890123456789012345678901234				

Response Code Text Table for Administrative Receipts

The denied administrative transaction response code text table is a separate table in the I3V7NAMS (for IBM 3624 version 7 terminals) or I3V8NAMS (for IBM 3624 version 8 terminals) and is used in conjunction with the #RA (Response Text Administrative) field type code when that particular field type code is used in an administrative receipt template for denied transactions.

The response code text table for denied administrative transactions (ADMIN^RESP^TEXT^G) provides additional receipt text for denied transactions. This text gives the service person an explanation for the denied transaction and is in addition to the text provided in the receipt template.

When the Device Handler process encounters a #RA field type code in a receipt template, the process takes the response code that it generated or that it received from the Authorization process, maps that code to an internal response code, moves to the administrative response code text table and scans for a match on the internal response code. From this point on, the Device Handler process uses the administrative response code text table in the same manner as the response code text table for financial transactions. This processing is described earlier in this section. The following entries are the default entries in the response code text table for denied administrative transactions. All of the default entries are in English; however, a second language can be defined for IBM 3624 version 8 terminals.

Response Code	Initial Message Text
16	UNABLE TO PROCESS AMT
29	INVALID AMT FOR HOPPER
33	ATM ALREADY BALANCED
39	ADMIN TRAN CANCELLED

Appendix A

IBM 3624 Terminal Servicing Procedures

This appendix outlines special servicing procedures performed by administrative personnel at IBM 3624 terminals. These procedures require the use of an administrative card.

Service operations for IBM 3624 terminals require special procedures when performed at the ATM. This appendix identifies those procedures and details the sequence in which those procedures must be performed in order to correctly update totals maintained by the ATM and keep BASE24 synchronized with the status of the ATM.

The following is a list of administrative procedures described in this appendix.

- Placing the terminal in service mode
- Servicing the terminal
- Printing machine subtotals
- Balancing the terminal
- Exiting service mode

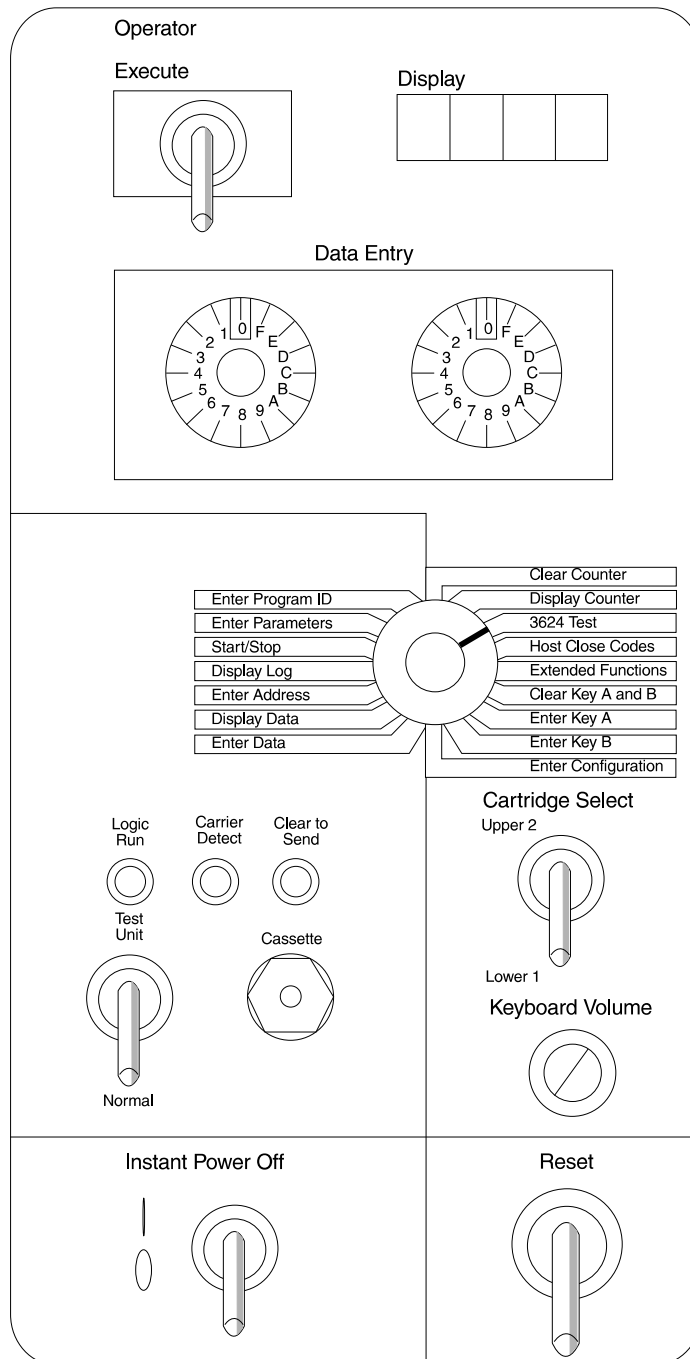
Administrative Access

To control on-site service access and security, special administrative cards are issued to institution personnel for the purpose of performing administrative functions. On-site, at the time of servicing the ATM, the service person must insert his or her administrative card into the ATM. The card information is then checked by BASE24, and if a positive match is made, the service person is granted administrative access to the ATM.

Note: Administrative card numbers are defined to BASE24 in the Cardholder Authorization File (CAF) or Negative Card File (NEG) depending on the type of authorization method used by the institution. If administrative cards are issued under a different prefix than the customer cards issued by the institution, the administrative card prefix must also be defined in the Card Prefix File (CPF).

IBM 3624 Service Control Panel

The following diagram depicts the service control panel of IBM 3624 terminals. This control panel is physically located in the rear compartment of the ATM and is used to perform a number of the servicing procedures described in this appendix.



Placing the Terminal in Service Mode

The following steps describe how to place the terminal in service mode. These steps must be performed prior to any other service procedures described in this appendix.

1. Insert your administrative card.
2. Enter your administrative PIN.
3. Press the Special key on the blue keyboard on the ATM (the key may be labeled “Special” or may be unlabeled).

This provides you with an Administrative Function Menu on the ATM screen.

4. Proceed to the next set of procedures, based on the actions you want to perform.
 - If you want to service the terminal (key 1 or 001 on the Administrative Function Menu depending the ATM version), refer to the topic “Servicing the Terminal.”
 - If you want to print current machine subtotals (key 2 or 002 on the Administrative Function Menu depending the ATM version), refer to the topic “Printing Machine Subtotals.”
 - If you want to balance the terminal (key 3 or 003 on the Administrative Function Menu depending the ATM version), refer to the topic “Balancing the Terminal.”
 - If you want to exit immediately, before performing any additional steps, press the red Cancel key.

Note: Do not press the Cancel key once you have started the next set of steps. From the Administrative Function Menu, if you press keys 1 or 3, you will need to continue with the respective procedure and exit service mode. Refer to the topic “Exiting Service Mode” for the steps necessary to exit the service mode and return the ATM to cardholder service operation.

Servicing the Terminal

The following activities are considered part of servicing the terminal and can be performed at the ATM under this procedure:

- Removing one or both money cartridges and replacing them with full cartridges of the same bill value
- Replenishing the statement/receipt paper supply
- Replacing printer ribbon
- Removing retained cards
- Collecting envelopes containing cardholder deposits
- Clearing counters on replenished money cartridges

These terminal servicing activities can be performed without affecting any of the information needed for terminal balancing. They need not be performed within the terminal balancing window. These procedures for servicing the terminal do not result in a receipt being produced.

The ATM is closed to cardholder use while the terminal is being serviced.

Procedures

The following steps describe how to service an ATM. In order to take these steps, the ATM must be in service mode (as described in the topic “Placing the Terminal in Service Mode”). These procedures begin from the Administrative Function Menu.

1. Press the 1 or 001 key depending on the ATM version.

This sets up the ATM for group 1 functions.

2. Remove your administrative card.

The following text should be visible on the CRT screen on the front of the ATM (for version 8 terminals):

```
THIS AUTOMATED TELLER MACHINE IS BEING  
SERVICED  
IT IS TEMPORARILY UNAVAILABLE FOR  
CUSTOMER TRANSACTIONS
```

This screen provides confirmation that a valid administrative request has been made. It is displayed until the ATM service representative exits service mode.

3. Proceed to the rear of the ATM; open the rear cabinet doors.
4. At the rear of the ATM, check and replenish the statement/receipt paper if necessary.
5. Verify the document transport mechanisms are free from jams and misfeeds; return the document transports to their service positions.

Note: You can now proceed to the topic “Exiting Service Mode” if this completes your service requirements.

6. Check the cash reject compartment.
7. Check and/or remove rejected cash according to institution policy. Ensure that any rejected cash is not jammed in the mechanism.
8. Return any inspected mechanisms to their service position.

Note: You can now proceed to the topic “Exiting Service Mode” if this completes your service requirements.

9. Replace any cash cartridge that has an insufficient amount of bills with a standard cash cartridge of the same bill value.
10. Clear the bill counter for each cartridge that has been replaced.
 - a. If cartridge 1 has been replaced with a standard cash cartridge, perform the following steps. Completing this procedure results in a cash increase for this cartridge.
 - 1) Set the Function Select switch on the service control panel to Clear Counters. (Refer to the illustration found previously in this appendix for a depiction of the service control panel.)
 - 2) Set the Cartridge Select switch to the Lower 1 position.
 - 3) Raise and release the Execute switch on the top of the service control panel to clear cartridge bill counter 1.
 - b. If cartridge 2 has been replaced with a standard cash cartridge, perform the following steps. Completing this procedure results in a cash increase for this cartridge.
 - 1) Set the Function Select switch on the service control panel to Clear Counters.
 - 2) Set the Cartridge Select switch to the Upper 2 position.
 - 3) Raise and release the Execute Switch on the top of the service control panel to clear cartridge bill counter 2.
11. Proceed to the topic “Exiting Service Mode.”

Printing Machine Subtotals

The following procedure describes how to print machine subtotals at the ATM. Machine subtotals reflect transactions that have occurred since the last balancing transaction. These totals are printed on an ATM receipt. Access to the rear of the ATM is unnecessary to complete this function.

Procedures

The following steps describe how to print machine subtotals at an ATM. In order to take these steps, the ATM must be in service mode (as described in the topic “Placing the Terminal in Service Mode”). These procedures begin from the Administrative Function Menu.

1. Press the 2 or 002 key depending on the ATM version.
2. Remove the receipt at the end of printing.

At the end of printing, the ATM returns automatically from service mode. If other administrative functions are to be performed, you must re-insert your card and place the terminal in service mode again. Refer to the topic “Placing the Terminal in Service Mode” for the procedures for placing the terminal in service mode.

Balancing the Terminal

The following activities are considered part of balancing the terminal and can be performed at the ATM under this procedure:

- Balancing the terminal
- Replenishing the statement/receipt paper supply
- Removing retained cards
- Emptying the rejected cash compartment
- Accounting for rejected cash
- Removing money cartridges
- Replacing money cartridges with the same bill value
- Clearing counters and resetting to zero for one or both money cartridges
- Updating standard cash balance

Warning: These balancing functions affect BASE24 files and can only be performed during the terminal balancing window if you have specified a balancing window. The TERMINAL BALANCE FLAG field on IDF screen 13 determines whether the terminal can be balanced only once a day during a window, or multiple times and at any time during the day. If this field contains the value 0, you are not allowed to balance the terminal outside of the window specified in the ATM BALANCE AND CUTOVER TIME WINDOW field on IDF screen 10.

The ATM is closed to cardholder use while the terminal is being balanced.

Procedures

The following steps describe how to balance an ATM. In order to take these steps, the ATM must be in service mode (refer to the topic “Placing the Terminal in Service Mode”). These procedures begin from the Administrative Function Menu.

1. Press the 3 or 003 key depending on the ATM version.

Receive an administrative receipt. The ATM is now balanced by the current cash method by BASE24.

2. Remove your administrative card.

The following text should be visible on the CRT screen on the front of the ATM (for version 8 terminals). This screen provides confirmation that a valid administrative request has been made. It is displayed until the ATM service representative exits service mode.

THIS AUTOMATED TELLER MACHINE IS BEING
SERVICED
IT IS TEMPORARILY UNAVAILABLE FOR
CUSTOMER TRANSACTIONS

3. Proceed to the rear of the ATM; open the rear cabinet doors.
4. At the rear of the ATM, check and replenish the statement/receipt paper if necessary.
5. Verify the document transport mechanisms are free from jams and misfeeds; return the document transports to their service positions.

Note: You can now proceed to the topic “Exiting Service Mode” if this completes your service requirements.

6. Check the cash reject compartment.
7. Check and/or remove rejected cash according to institution policy. Ensure that any rejected cash is not jammed in the mechanism.
8. Return any inspected mechanisms to their service position.

Note: You can now proceed to the topic “Exiting Service Mode” if this completes your service requirements.

9. Replace any cash cartridge that has an insufficient amount of bills with a standard cash cartridge of the same bill value.
10. Clear bill counter for any cartridge that has been replaced.
 - a. If cartridge 1 has been replaced with a standard cash cartridge, perform the following steps. A standard cash balance results from the difference between standard cash and ending cash.
 - 1) Set the Function Select switch on the service control panel to Clear Counters. (Refer to the illustration found previously in this appendix for a depiction of the service control panel.)
 - 2) Set the Cartridge Select switch to Lower 1.
 - 3) Raise and release the Execute switch on the top of the service control panel to clear cartridge bill counter 1.

- b. If cartridge 2 has been replaced with a standard cash cartridge, perform the following steps. A standard cash balance results from the difference between standard cash and ending cash.
 - 1) Set the Function Select switch located on the service control panel to Clear Counters.
 - 2) Set the Cartridge Select switch to Upper 2.
 - 3) Raise and release the Execute switch on the top of the service control panel to clear cartridge bill counter 2.
- 11. Check the cash reject compartment.
- 12. Replace and/or remove any rejected cash according to institution policy.
- 13. Verify that any rejected cash is not jammed in the mechanism.
- 14. Check and replenish statement/receipt paper if needed; restore document transport mechanisms to service positions.
- 15. Proceed to the topic "Exiting Service Mode."

Exiting Service Mode

The following steps describe how to exit service mode and return the ATM to cardholder service operation. Exiting service mode causes the ATM to perform various tests to verify that its systems are operational and ready for cardholder use.

These procedures are all performed from the service control panel (refer to the illustration found previously in this appendix for a depiction of the service control panel).

1. Turn the Function Select switch to the 3624 Test position.
2. Raise and release the Execute switch.

When the display window shows a value of C0C0, the ATM is ready to resume cardholder transactions. If the value C0C0 does not appear in the display window, inspect transport mechanisms for correct service positions; then repeat the steps of the exit procedure. If values other than C0C0 still appear, there is a malfunction and the ATM will remain out of service for cardholder use. Refer to the vendor documentation for information about the proper procedures to follow in the case of a malfunction.

3. Turn the Function Select switch to the Extended Functions position.
4. Set the data entry dials to the values of A and 1.
5. Raise and release the Execute switch.

The control panel display should show A1.

6. Set the data entry dials to the values of F and E.
7. Raise and release the Execute switch.

The values EC90 and 00A1 should appear on the control panel display. If these values do not display, you may be able to reopen the terminal from the DCT. Refer to the vendor documentation for information about the proper procedures to follow in the case of a malfunction.

8. Close and secure the service door on the back of the ATM only after the value 00A1 displays.

Warning: Do not close the rear service door on the ATM before you see the value 00A1 on the display panel. Closing the door prior to the 00A1 display can disrupt the Device Handler process sequencing of the ATM.

After the exiting procedure has completed, no service activities can be performed without placing the terminal in service mode again.

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